Analytical Cume (Feb. 9, 2007)
Topic: Chromatography

1. (10 points) Three components, A, B, and C, of a mixture are separated on a given chromatography column. The partition coefficient (or distribution constant), K, for each of these species is: $K_A = 56$, $K_B = 100$, $K_C = 1$. In what order do you expect A, B, and C to elute off of the column, from first to last? Explain why.

2. (30 points, 6 points each) In column chromatography, how could you change each of the following parameters to improve chromatographic efficiency? (Examples of responses: "Increase", "decrease", "change will not have an effect", or "change will have any effect under a given set of conditions") Briefly explain why.
(Consider each parameter independently of the others. Do not assume that you will be changing all of them simultaneously.)

(a) flow rate
(b) diameter of particles in a packed column
(c) diameter of a tubular column
(d) temperature (assuming it is a gas chromatography separation)
(e) thickness of the stationary phase (assuming that a viscous liquid coats the particles of a packed column or coats the wall of a tubular column)

3. (15 points, 5 points each) What is the name of the chromatography method that is based on the following chemical interactions or features? (Just write the name… do not explain anything).

(a) charge
(b) hydrophilicity/hydrophobicity
(c) specific binding to antibodies
(d) molecular weight or size
(e) adsorptive properties to surfaces

4. (45 points) The following data were obtained by running GC on a fused silica column of an artificial mixture of cinnamon flavor. There are six essential oil components, one of which is cinnamaldehyde, which you might recognize. Two of the six essential oil components are listed below. The dimensions of the column were **0.31 mm diameter x 35 cm length with a 0.25-μm film** stationary phase.

compound	t_R, min	w, min
air	0.82	----
thymol	19.6	0.30
carvacrol	19.9	0.40

Calculate:

(a) **(8 points)** the AVERAGE number of plates from the data
(b) **(7 points)** the average plate height for the column
(c) **(8 points)** Calculate the resolution for the two compounds
(d) **(7 points)** What is the capacity factor (retention factor) for each of the solutes?
(e) **(7 points)** What is the ratio of the equilibrium constant between the stationary and mobile phases of the slower moving species to the equilibrium constant for the faster moving species?
(f) **(8 points)** If the length of the column is increased in order to obtain a resolution of 1.5, what would be the new retention time for the slower moving species?

Equations, Constants, and Tables that May be of Use on this Exam

$$K_A = \frac{C_{s,A}}{C_{m,A}}$$

$$k'_A = \frac{moles_s}{moles_m} = \frac{C_s V_s}{C_m V_m} = K_A \frac{V_s}{V_m} = \frac{(t_{r,A} - t_m)}{t_m}$$

$$\alpha = \frac{t_{r,B} - t_m}{t_{r,A} - t_m} = \frac{t'_{r,B}}{t'_{r,A}} = \frac{k'_B}{k'_A} = \frac{K_B}{K_A} \quad \text{where } t'_{r,B} > t'_{r,A}$$

$$R_s = \frac{2(t_{r,B} - t_{r,A})}{w_A + w_B} = \frac{\Delta t_r}{w_{av}}$$

$$H \cong A + \frac{B}{u} + C\,u$$

$$H \cong \frac{B}{u} + C_s\,u + C_m\,u$$

$$\sigma^2 = H\,L$$

$$N = \frac{L}{H}$$

$$N = \frac{16\,t_r^{\,2}}{w^2}$$

$$R_s = \frac{\sqrt{N}}{4}\left(\frac{\alpha - 1}{\alpha}\right)\left(\frac{k'_b}{1 + k'_{av}}\right)$$

$$t_{r,B} = \frac{16\,R_s^2\,H}{u}\left(\frac{\alpha}{\alpha - 1}\right)^2\left(\frac{(1 + k'_b)^3}{(k'_b)^2}\right)$$